Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Cloud-focused software engineer with 2+ years using **AWS**, **Python**, SQL, Terraform, Docker, and CI/CD. Experienced automating cloud data workflows, improving reliability and cost efficiency, and documenting production systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Database Systems, Software Engineering, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Data Retrieval, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected multi-tenant cloud storage using Azure SQL and Blob Storage for 500+ users, applying row-level security, normalized schemas, indexing, and query optimization to reduce dashboard latency 71%.
- Built ETL integrations across FastAPI, Supabase, SQL Server, and cloud storage, processing 5,000+ monthly jobs with validation, reconciliation, error handling, documentation, and CI/CD-backed releases achieving 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Implemented an AWS-hosted Python backend serving 150+ instructors and 10,000+ daily requests, using AWS RDS, REST APIs, and CI/CD while maintaining stable application performance during peak workloads.
- Designed concurrent batch processing and secure PDF, DOCX, and ZIP ingestion with validation, failure isolation, model-usage tracking, and automated workflows, reducing instructor feedback turnaround 60% in production.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*

- Automated cross-platform builds and 20+ unit and integration suites through Jenkins, PyTest, Mockito, and VM-based pipelines, validating releases across Windows, Linux, and macOS with documented failure reports.
- Optimized API and ETL integrations using Python, AWS Glue, S3, SQL, indexing, and caching, improving response times 30%, increasing pipeline efficiency 68%, and reducing cloud storage costs 15%.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*

- Developed a secure microservice-based resource platform synchronizing two million records through REST APIs and WebSockets, supporting authentication, customer, billing, permissions, and notification workflows across distributed application services.
- Automated Docker and Jenkins deployment workflows, cutting release time 50% while consistently maintaining 95%+ successful build validation and using operational analytics to reduce resource waste by 42%.

Skills

Cloud & Infrastructure: AWS, EC2, S3, RDS, Glue, ECS, Fargate, CodePipeline, Terraform, Azure, GCP, Docker, Kubernetes
AWS Architecture: IAM, VPC concepts, private subnets, Route 53, load balancing, networking, tagging, budgets, cost optimization, Well-Architected principles
Languages & Data: Python, SQL, Java, JavaScript, TypeScript, C++, Bash, PostgreSQL, MySQL, SQL Server, MongoDB, Firebase
Automation & Operations: Jenkins, GitHub Actions, CI/CD, infrastructure as code, PyTest, Mockito, Git, Linux, monitoring, logging, incident troubleshooting
Backend & Integration: FastAPI, Flask, Node.js, REST APIs, WebSockets, ETL, data governance, schema design, caching, API integration
Practices: Cloud security, access control, cost forecasting, Agile, documentation, code reviews, system design, production support, stakeholder communication

Projects

Civic Cloud Guardrails | *AWS, Terraform, Python, IAM, VPC, Budgets, PyTest*

- Built a deployment-safe AWS governance reference with Terraform-defined private networking, scoped IAM, encrypted audit storage, mandatory tags, and workload budgets plus Python forecasting validated by three automated tests.

Linguistix | *Python, AWS ECS/Fargate, CodePipeline, Docker, CI/CD*

- Developed and deployed a containerized Python data-processing API on AWS ECS/Fargate with CodePipeline automation and validation checks, achieving 87% processing accuracy while maintaining reliable, repeatable cloud releases.

Chat2SQL | Code | *Java, TypeScript, SQL, Docker*

- Built a Docker-based SQL development workspace supporting 10+ database systems, schema exploration, structured validation, testing, and cross-platform deployment through Java, TypeScript, Maven, Node.js, and reproducible local environments.